Praneel Magapu

(919) 559-4602 | pmagapu5@gmail.com | linkedin.com/in/praneel-magapu | github.com/mpraneel

EDUCATION

North Carolina State University Raleigh, NC
Bachelor of Science in Computer Science — GPA: 3.4/4.0 *Expected Dec 2026*
Relevant Coursework: Data Structures & Algorithms, Systems Programming, Software Engineering, AI/ML Concepts

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Bash
Frameworks / Libraries: Flask, FastAPI, React, Tailwind CSS, NLTK, Pydantic
Tools / Platforms: Git, Docker, AWS (Elastic Beanstalk, CLI), Firebase, Supabase, Google Apps Script, Jenkins, GitHub Actions
AI & Data: LLM integration, RAG systems, Multimodal AI (TwelveLabs), NLP (Regex, NLTK), MongoDB, SQL

EXPERIENCE

DevDynamics.ai Remote
Software Engineering Intern *May – Aug 2025*

- Engineered LLM-powered intent extraction pipeline and automated processing workflows, reducing manual triage overhead across engineering teams.
- Built a **hybrid regex + LLM classifier** with Google Workspace APIs to automatically classify and track email thread timelines across **100+ active threads**.
- Deployed production **AI-powered Gmail Auto-Labeler** using Flask, AWS, and Supabase; adopted internally for day-to-day email management.

Liquid Rocketry Lab — NC State Raleigh, NC
Data Engineer *Sep 2024 – May 2025*

- Built **real-time telemetry pipeline** for live rocket engine tests using Python and MongoDB across containerized microservices, capturing abort and breach events with sub-second latency.
- Implemented observable fault-detection pattern for automated abort detection and breach logging, improving safety response reliability during high-frequency sensor ingestion.

North Carolina State University Raleigh, NC
Research Assistant *May 2024 – Dec 2025*

- **Game2Learn Lab (Dr. Tiffany Barnes):** Built production RAG system powering MerryQuery, a generative AI educational assistant; drove full-stack integration and iterated on UI/UX from live pilot feedback.
- **Crowd Label Quality Control (Dr. Ed Gehringer):** Extended algorithms to detect unreliable crowd labels at scale using interval analysis and iterative aggregation, improving integrity of large ML training datasets.

PROJECTS

Mendacia — Python, Flask, React, TypeScript, TwelveLabs API, LLM APIs, Tailwind CSS *Hack_NCState 2026*

- Built a **full-stack multimodal media forensics platform** generating structured forensic reports from raw video, detecting propaganda, narrative distortion, and cross-modal inconsistencies using TwelveLabs scene intelligence and LLM reasoning.
- Designed a **cross-modal consistency engine** comparing narrative claims against visual scene data with timestamped reasoning, and a defensive adapter layer enforcing a stable UI contract across volatile backend responses.

Hiring Agent — FastAPI, Pydantic, React, Docker, pytest, GitHub Actions *Jul – Aug 2025*

- Built end-to-end **resume-to-job matching engine** with ontology-based skill normalization and **hybrid keyword + LLM scoring** with rationale generation; exposed via REST API with structured Pydantic contracts.
- Containerized with **Docker** and validated via **pytest + GitHub Actions** CI pipeline; designed for extensibility across job board integrations.

Mesh Alignment Engine — Python, Open3D, NumPy, Trimesh, Matplotlib *Mar – May 2026*

- Implemented **ICP-based surface mesh registration pipeline** aligning 3D scans with sub-millimeter RMSE; applied multi-resolution coarse-to-fine strategy to accelerate convergence on high-density meshes.
- Built **deviation heatmap visualization** mapping per-point geometric divergence between aligned meshes, enabling spatial quality analysis analogous to treatment tracking in orthodontic scan pipelines.